ML Lab

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Ensemble Methods

Ensemble Classification — Bagging and Boosting

Why Ensemble Learning?

- **Instead of relying on a single classifier, combine multiple classifiers to obtain a stronger model.**
- **Why single models fail?**
  - High bias → underfitting (e.g., linear models)
  - High variance → overfitting (e.g., decision trees)
  - Sensitive to noise and sampling variation

  📌 Ensemble methods aim to reduce error by aggregation.
- **What Is an Ensemble?**

An ensemble classifier consists of:

- A set of base learners: $h_1(x), h_2(x), \ldots, h_T(x)$
- A combination rule (voting / weighted sum)

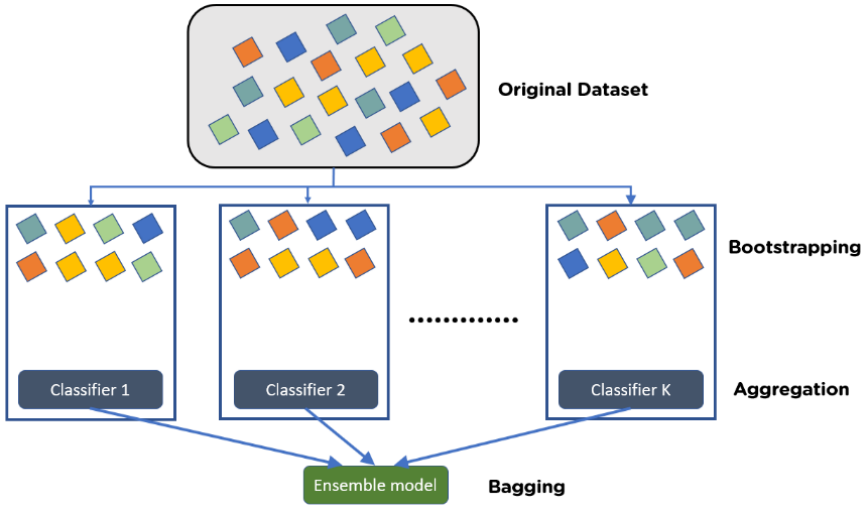
Final prediction (classification) $\hat{y} = \text{majority vote}\{h_1(x), \ldots, h_T(x)\}$

Types of Ensemble Methods

Method	Main Goal	Learners
Bagging	Reduce variance	Independent
Boosting	Reduce bias	Sequential
Stacking	Combine diversity	Heterogeneous

Bagging (Bootstrap Aggregating)

- **Bagging is an ensemble technique that reduces variance by training many models independently on different bootstrap samples of the same dataset and aggregating their predictions.**
- **Core Idea (Intuition)**
  - Resample the data many times (with replacement).
  - Train one model per sample (all in parallel).
  - Aggregate predictions (vote/average).
- **Why Bagging is Needed (Motivation)**
  - Some learning algorithms (especially decision trees) are high-variance
  - Small changes in training data → very different models
  - Leads to overfitting
- **Goal of Bagging:** Reduce variance without increasing bias
- **Why Bagging Reduces Variance:** $\mathrm{Var}\left(\frac{1}{T}\sum h_t(x)\right) < \mathrm{Var}(h_t(x))$



When Bagging Works Best

- ✔ High-variance models
- ✔ Unstable learners

Random Forest = Bagging + Feature Randomness

- **What bagging does**
  - Bagging uses bootstrap sampling
  - Each tree is trained on a different resampled version of the dataset
  - Sampling is with replacement
- **What feature randomness does**
  - At each split:
    - Instead of considering all features
    - Random Forest considers only a random subset of features

Random Forest is defined as:

$$\boxed{\text{Random Forest} = \text{Bootstrap sampling} + \text{Random feature selection}}$$

Where:

- Bootstrap sampling → **data-level randomness**
- Random feature selection → **feature-level randomness**